№1

$$|M|(E) = \sup \sum_{i=1}^{\infty} |M(E_i)|$$

1) Пусть $E_i = \emptyset$ при $i > N$:

$$\sup \sum_{i=1}^{N} |M(E_i)| = \sup \sum_{i=1}^{\infty} |M(E_i)| \leq |M|(E) \Rightarrow$$

$$\Rightarrow \lambda(E) \leq |M|(E)$$

2) $\forall \varepsilon > 0 \ \exists \{A_i\}_{i=1}^{\infty} : \sum_{i=1}^{\infty} |M(A_i)| > |M|(E) - \frac{\varepsilon}{2}$

$$\sum_{i=1}^{\infty} |M(A_i)| - \text{сх.} \Rightarrow \exists N : \sum_{i=N+1}^{\infty} |M(A_i)| < \frac{\varepsilon}{2}$$

Пусть $E_1 = A_1, \ldots, E_N = A_N, E_{N+1} = \bigcup_{i=N+1}^{\infty} A_i$

$$\sum_{i=1}^{N+1} |M(E_i)| = \sum_{i=1}^{N} |M(A_i)| + |M(\bigcup_{i=N+1}^{\infty} A_i)|$$

$$\sum_{i=1}^{N+1} |M(E_i)| \geq \sum_{i=1}^{N} |M(A_i)| = \sum_{i=1}^{\infty} |M(A_i)| - \sum_{i=N+1}^{\infty} |M(A_i)| >$$

$$> (|M|(E) - \frac{\varepsilon}{2}) - \frac{\varepsilon}{2} = |M|(E) - \varepsilon$$

$$\lambda(E) \geq \sum_{i=1}^{N+1} |M(E_i)| > |M|(E) - \varepsilon$$

$$\lambda(E) \geq |M|(E)$$

№2

$$\#(A) = 0 \Rightarrow A = \emptyset \Rightarrow dx(A) = 0$$

$\#$ не имеет разложения отн. dx

№3

$$f \in L^p([0,1]) \Leftrightarrow \int_0^1 |f(x)|^p dx < \infty$$

$$f \in L^{\infty}([0,1]) \Leftrightarrow \sup_{x \in [0,1]} |f(x)| < \infty \text{ п.в.}$$

$$\int_0^1 |f(x)|^p dx \le \int_0^1 M^p dx \le M^p \quad \forall p : 1 \le p < \infty$$

$$\Rightarrow f \in L^p([0,1])$$

$$f = \ln x$$

$$\int_0^1 |\ln x|^p dx = \left| \begin{array}{l} x = e^{-t} \\ dx = -e^{-t}dt \\ x(0) = +\infty \\ x(1) = 0 \end{array} \right| = \int_0^{+\infty} t^p e^{-t} dt = \int_0^1 t^p e^{-t} dt + \int_1^{+\infty} t^p e^{-t} dt$$

$$0 \le t^p e^{-t} \le 1 \text{ при } t \in [0,1]$$

$$t^p \le e^{t/2} \text{ при } t \to +\infty \Rightarrow t^p e^{-t} \le e^{-t/2}$$

$$\int_1^{\infty} t^p e^{-t} dt \le \int_1^{\infty} e^{-t/2} dt = 2e^{-1/2} < \infty$$

$$\Rightarrow f \in \bigcup_{1 \le p < \infty} L^p([0,1]), \text{ но } f \notin L^\infty([0,1])$$

$$\Rightarrow L^\infty([0,1]) \subset \bigcup_{1 \le p < \infty} L^p([0,1])$$

№4

$$\lambda \ll \mu \Leftrightarrow \forall E \in \mathcal{M} : \mu(E) = 0 \Rightarrow \lambda(E) = 0$$

$$\forall E = \bigcup E_i : \mu(E) = 0 \Rightarrow \mu(E_i) = 0 \Rightarrow \lambda(E_i) = 0 \Rightarrow$$

$$\Rightarrow |\lambda|(E) = 0 \Rightarrow \text{если } \lambda \ll \mu, \text{ то } |\lambda| \ll \mu. \text{ Пусть } \lambda \ll \mu$$

(!): $\forall \varepsilon > 0 \ \exists \delta > 0 : \mu(E) < \delta \Rightarrow |\lambda|(E) < \varepsilon$
Пусть $\exists \varepsilon > 0 \ \forall \delta > 0 \ \exists E \in \mathcal{M} : \mu(E) < \delta, \ |\lambda|(E) \ge \varepsilon$

$$\{E_n\}_{n=1}^{\infty} : \delta = 2^{-n} \Rightarrow \mu(E_n) < 2^{-n}, \ |\lambda|(E_n) \ge \varepsilon$$

$$\sum_{n=1}^{\infty} \mu(E_n) < \infty \Rightarrow \mu(\limsup_{n\to\infty} E_n) = 0 \Rightarrow |\lambda|(\limsup_{n\to\infty} E_n) = 0$$

$$|\lambda|(\bigcup_{k \ge n} E_k) \ge |\lambda|(E_n) \ge \varepsilon$$

$$|\lambda|(\limsup_{n\to\infty} E_n) = |\lambda|(\bigcap_{n=1}^{\infty} \bigcup_{k \ge n} E_k) = \lim_{n\to\infty} |\lambda|(\bigcup_{k \ge n} E_k) \ge \varepsilon$$

св-во непрерывности $\Rightarrow$ противоречие. (Теорема Бореля-Кантелли)

2) Пусть $\mu(E) = 0 \Rightarrow \forall \varepsilon > 0 : \mu(E) < \delta \Rightarrow |\lambda|(E) < \varepsilon \Rightarrow |\lambda|(E) = 0 \Rightarrow \lambda \ll \mu$

№5

$z_n \in A$; $|S - z_n| \le |S|$, где $S = \sum_{n\in A} z_n$

$$|S - z_n|^2 \le |S|^2$$

$|S - z_n|^2 = (S - z_n)(\overline{S - z_n}) = (S - z_n)(\overline{S} + \overline{z_n}) = |S|^2 - 2\,\mathrm{Re}(S\overline{z_n}) + 1 \le |S|^2$

(здесь предполагается, что $|z_n|^2 = 1$)

$$2\,\mathrm{Re}(S\overline{z_n}) - 1 \ge 0 \Rightarrow \mathrm{Re}(S\overline{z_n}) \ge \frac{1}{2} > 0$$

$$u = \frac{S}{|S|},\ |u| = 1 \Rightarrow \mathrm{Re}(z_n\overline{u}) = \mathrm{Re}(u\overline{z_n}) = \frac{1}{|S|}\,\mathrm{Re}(S\overline{z_n}) > 0$$

$$\Rightarrow \forall n \in A : \mathrm{Re}(z_n\overline{u}) > 0 \Rightarrow \forall n \in A : \mathrm{sgn}(\mathrm{Re}\, w_n) \ge 0 \text{ или } \le 0$$

№7

$$||v_n - v_m||^2 = \int_0^1 |v_n' - v_m'|^2 dx \to 0,\ \text{но фундаментальная}$$

$$L^2([0,1]) \text{ — полное } \Rightarrow \exists g \in L^2([0,1]) : v_n' \to g$$

$$u(x) = \int_0^x g(t)dt; \quad v_n(x) = \int_0^x v_n'(t)dt$$

$$v_n(x) \to u(x); \quad u(0) = 0,\ u(1) = 0$$

Элементы пополнения: $u \in L^2([0,1])$, $u' \in L^2([0,1])$, $u(0) = u(1) = 0$

(!): $||u||_{L^2} \le C||u||_W$

Нер-во Коши-Буняковского: $(\int_0^x f(t)g(t)dt)^2 \le (\int_0^x f^2(t)dt)(\int_0^x g^2(t)dt)$

$$|v(x)|^2 = |\int_0^x v'(t)\cdot 1 dt|^2 \le (\int_0^x 1^2 dt)(\int_0^x (v'(t))^2 dt) = x\int_0^x (v'(t))^2 dt \le x\int_0^1 (v'(t))^2 dt$$

$$\int_0^1 |v(x)|^2 dx \le \left(\int_0^1 (v'(t))^2 dt\right)\left(\int_0^1 x dx\right) = \frac{1}{2}\int_0^1 (v'(t))^2 dt$$

$$\Rightarrow ||v||_{L^2}^2 \le \frac{1}{2}||v'||_{L^2}^2$$

№7 (продолжение)

$$F \in L^2([0,1])$$

$$L_F(v) = \int_0^1 F(x)v(x)dx, \quad \forall v \in W$$

$$||v||_{L^2} \leq \frac{1}{\sqrt{2}}||v||_W$$

$$|L_F(v)| \leq ||F||_{L^2}||v||_{L^2} \leq \frac{1}{\sqrt{2}}||F||_{L^2}||v||_W \Rightarrow$$

$$\Rightarrow L_F \text{ — непр. функционал, линейный}$$

Теорема Рисса: W — гильбертово, L — непр. линейный функционал $\Rightarrow \exists! u \in W$:
$L(v) = \langle u, v \rangle_W, \quad \forall v \in W$

$$\Rightarrow \forall v \in W \ \exists! u \in W : \int_0^1 u'(x)v'(x)dx = \int_0^1 F(x)v(x)dx$$

$$\int_0^1 F(x)v(x)dx = -\int_0^1 u''(x)v(x)dx$$

$$\int_0^1 (-u''(x) - F(x))v(x)dx = 0 \Rightarrow F(x) = -u''(x)$$